A COMMUNITY IN FORMATION: BUILDING AND ANALYSING A CORPUS OF SLOVENE POSTS ON BLUESKY

Nives HÜLL[1]

[1]University of Ljubljana, Ljubljana, Slovenia

This paper presents a corpus of 141,013 public Slovene-language posts by 432 authors on Bluesky, spanning August 2023 to mid-April 2026. The corpus was built using a protocol-native ATProto workflow combining relay-based discovery of Slovene-tagged posts, DID-resolved backfill of surviving public post histories from authors' Personal Data Servers, and post-level language filtering validated through manual annotation. As one of the first language-specific empirical studies of Bluesky use for a small European language community, the corpus documents the emergence of a new public space on an open-protocol platform. Based on the collected data, a marked increase in posting activity is visible in late 2024. Across the collection period, reply posts account for a large share of the material (68.3%), including in the months before this increase, indicating that conversational use was already present prior to the surge. Further observed characteristics include sparse and largely account-concentrated hashtag use, and content sharing shaped primarily by Slovene news media, GIF-based informal exchange, and Bluesky-native tooling. The paper illustrates how protocol-level access in ATProto enables the construction of a high-precision corpus for an under-resourced language without relying on platform-specific research APIs. At the same time, it raises legal and ethical questions that arise when small and highly identifiable language communities become reconstructable from public, decentralised infrastructures. The paper documents the collection workflow, its validation, and the basic temporal, interactional, and linking properties of Slovene-language activity on Bluesky, offering a reproducible case study for digital humanities research on open-protocol social media.

Keywords: Bluesky, ATProto, Slovene, corpus construction, social media

1 INTRODUCTION

Bluesky and the wider ATProto ecosystem have become increasingly relevant as a source of public social-media data (Failla & Rossetti, 2024; Quelle & Bovet, 2025). For smaller languages such as Slovene, this opens up an opportunity to document an emerging online community on a new, decentralised platform, and to consider how such documentation should be done. The paper is also motivated by the author's

long-standing interest in Slovene online communities, by familiarity with the emerging Slovene Bluesky space, and by a broader interest in what open-protocol social media makes visible and researchable.

This paper describes an attempt to build a Slovene Bluesky corpus from public data. The main challenges are not only in collection, but in deciding what should count as a Slovene post. Language tags are useful but imperfect, and posts by Slovene-linked users are not automatically Slovene. The approach combines protocol-aware collection with explicit validation, but it still requires drawing a practical line around what counts as a Slovene post.

Four questions guide the work. The first is whether a usable Slovene Bluesky corpus can be built from public ATProto data. The second concerns how far collection can move beyond a simple `langs=sl` strategy. The third asks which inclusion rules are reliable enough for a main corpus. The fourth asks what basic properties are visible in the resulting corpus.

This paper makes two contributions. First, it presents a protocol-native workflow for constructing a high-precision corpus of Slovene Bluesky posts from public ATProto data, combining tagged discovery, multi-PDS backfill, and validated post-level language filtering. Second, it uses the resulting corpus as an empirical case study of a small online community in formation. The analysis shows a sharp late-2024 growth surge, strongly conversational posting patterns, and a highly concentrated participation structure, together suggesting that the Slovene Bluesky sphere is emerging as a small conversational public space.

2 RELATED WORK

2.1 Slovene language resources and social-media corpora

Written standard Slovene is well served by reference resources. Gigafida 2.0 (Krek et al., 2020a; Krek et al., 2019) is the main reference corpus, with about 1.1 billion words of newspaper, web, and book text; ssj500k (Krek et al., 2020b) provides manually annotated training data; siParl 3.0 (Pančur et al., 2022) covers parliamentary proceedings from 1990 to 2022. The broader methodological context is surveyed in Gorjanc and Krek (2005).

The most directly relevant prior work is the JANES project (*Jezikovna analiza nestandardne slovenščine*), a central Slovene resource for computer-mediated communication, drawing primarily on Twitter but also covering news comments and forum posts (Erjavec et al., 2019). JANES mapped out much of the territory: the feasibility

of building Slovene social-media corpora and the specific challenges they pose, including non-standard spelling, code-switching, short texts, and unreliable platform language tags. The present work is much smaller in scale and ambition; it revisits some of the same questions in the context of Bluesky.

The closest language-specific Bluesky precedent located so far is Tjong Kim Sang (2025), who evaluates Bluesky as a successor data source for Dutch social-media research after Twitter/X API restrictions made the TwiNL Twitter archive difficult to extend. That study collected 4.2 million Dutch Bluesky posts from November 2024 to May 2025 through keyword-based search and demonstrates compact corpus-linguistic analyses such as word-frequency comparison, daily rhythms, event response, hashtags, and political vocabulary. The present paper is narrower in language-community size, but differs methodologically by using protocol-level ATProto backfill from discovered authors and by making post-level language validation a central part of corpus construction.

2.2 Bluesky and ATProto

Prior work characterises Bluesky as both Twitter-like and decentralised, and links parts of its early growth to events around Twitter/X (Failla & Rossetti, 2024; Quelle & Bovet, 2025). Platform access conditions also matter here: Bluesky operated as an invite-only beta through 2023, regulating growth through waitlists and invite codes (Bluesky Social PBC, 2023b, 2023c), and only opened to the public on 6 February 2024, when invites were removed (Bluesky Social PBC, 2024). The present corpus cannot establish why individual Slovene users joined Bluesky, but the late-2024 surge visible in Section 5 should be interpreted against this broader platform context: public opening had already removed one barrier to entry, while the Slovene corpus shows its main surge only later, suggesting that public opening was a background enabling condition rather than a sufficient explanation on its own.

Technically, Bluesky is built on the AT Protocol (ATProto), a federated specification that separates identity from hosting: user identities are DIDs (decentralised identifiers), and public records are stored in per-account repositories on Personal Data Servers (Bluesky Social PBC, n.d.-a, 2023a; Kleppmann et al., 2024). ATProto also provides public synchronisation mechanisms: repository exports and a repository event stream, commonly called the firehose (Bluesky Social PBC, n.d.-b, n.d.-c). From a corpus-collection standpoint, this means that public posts can be collected through protocol-level mechanisms rather than through a separate academic access programme. DID resolution allows a collector to follow an account regardless of which PDS hosts it, a property exploited by the multi-PDS collection approach described in this paper.

Existing empirical work on Bluesky has mostly operated at large scale and focused on platform-wide, network-level, or topic-specific phenomena. Failla and Rossetti (2024) assembled a dataset of about 235 million posts from over four million users for behavioural analysis and reported large language-tag clusters, especially English, Japanese, and German. Quelle and Bovet (2025) used Bluesky data to characterise network topology, political polarisation, and user-generated feeds. More recent work treats Bluesky's discovery mechanisms as research objects in their own right: Balduf et al. (2025) analyse starter packs as a mechanism for social bootstrapping, while Hanny et al. (2026) use multilingual keyword-based Bluesky collection for disaster analysis. These studies make clear that Bluesky is not simply a Twitter replacement: feeds, starter packs, repository access, and language metadata all affect what can be collected and interpreted.

Language-specific Bluesky corpus work remains sparse. The Dutch study by Tjong Kim Sang (2025) is therefore a useful comparison point, while this paper instead offers a smaller but more validation-oriented case study for Slovene and for ATProto-based collection of a small language community.

2.3 Language identification in corpus construction

Building a language-specific corpus from a multilingual stream typically combines two signals: platform-supplied language metadata and automatic language identification. Neither is reliable on its own. Bluesky posts may contain a `langs` field with one or more BCP-47 language codes, and the official client auto-detects post languages while allowing users to override them (Bluesky PBC, n.d.-b). Large-scale Bluesky studies have also found that language metadata can be inconsistent or incomplete: Failla and Rossetti (2024) report non-standard and ill-formed language tags, while Hanny et al. (2026) caution that language metadata was inaccurate enough in a multilingual disaster crawl that filtering only by existing tags would risk missing relevant posts.

In the present corpus, 41.4% of posts carry both `sl` and `en` tags, while langid assigns English to only 278 posts. This mismatch is one reason to treat platform language tags as useful metadata rather than ground truth. Automatic language identification tools—langid.py (Lui & Baldwin, 2012) and langdetect (Nakatani, 2010) are used here—are also imperfect, especially for short texts and closely related South Slavic languages. Using consensus across multiple detectors and requiring agreement with author-supplied tags improves precision, which is the rationale for the tiered inclusion strategy described in Section 3.

3 DATA COLLECTION AND METHODOLOGY

The corpus was built through a layered pipeline: tagged discovery through public relay mechanisms, seed-author identification, backfill from each author's home server, and post-level language filtering. Each phase is described below.

3.1 Phase 1: Tagged discovery

Two complementary collection methods were used to discover Slovene-linked posts through public Bluesky relay infrastructure.

First, a **historical crawl** queried the Bluesky collection directory for public repositories containing `app.bsky.feed.post` records. The relevant relay endpoint is `listReposByCollection` (Bluesky PBC, n.d.-a; Bluesky Social PBC, n.d.-c). For each discovered repository, the author's DID was resolved to its home Personal Data Server (PDS), and the post records were fetched via `listRecords` from that PDS. Only posts whose `langs` field contained the tag `sl` or any `sl-*` subtag were retained.

Second, a **live firehose** subscriber connected to the public repository event stream (`subscribeRepos` on `wss://bsky.network`) and processed post creation, update, and deletion events in real time, again retaining only Slovene-tagged posts (Bluesky Social PBC, n.d.-b). The firehose subscriber also handled account-level events (deletions, deactivations, suspensions) by marking the affected posts accordingly in the local store.

Both sources stored their results in local SQLite databases, with posts keyed by their AT-URI and soft-deletion tracking for posts that were later removed by their authors.

3.2 Phase 2: Seed author identification

Every unique author DID that appeared in at least one Slovene-tagged post from either discovery source was collected into a seed author list. The resulting list defines the author population for the rest of the pipeline. The corpus is therefore not a sample of the Bluesky post stream, but an author-centred collection of currently recoverable posts from a defined set of discovered Slovene-linked accounts.

3.3 Phase 3: Author-history backfill

For each seed author, the available surviving public post history was retrieved through ATProto repository access. Each author's DID was resolved through a public directory: `plc.directory` for `did:plc:` identifiers, or the `.well-known/did.json` endpoint

for `did:web:` identifiers. The resolved PDS URL was cached locally to avoid repeated lookups. Posts were then fetched via paginated `listRecords` calls to the author's home PDS.

Because resolution goes through the DID rather than through a fixed domain, the backfill was not restricted to accounts on `bsky.social`; accounts hosted on any public ATProto PDS were handled identically. This phase retrieved surviving public posts from each seed author, regardless of language tag, transforming the corpus from a tag-filtered post stream into an author-centred collection. Deleted posts that were no longer available at crawl time are not recoverable through this method and are out of scope.

The resulting expanded store contained 379,482 posts from 742 active authors, spanning the earliest surviving post (May 2023) through the collection date (April 2026).

3.4 Phase 4: Post-level language filtering

Each post in the expanded store was then classified individually to determine whether it should be included in the Slovene corpus. The classifier applied three binary signals:

- **Tag signal**: does the post's `langs` field contain `sl` or `sl-*`?
- **langid signal**: does the langid.py language identifier (Lui & Baldwin, 2012) assign the label `sl` to the post text?
- **langdetect signal**: does the langdetect library (Nakatani, 2010) assign Slovene a probability of at least 0.90?

Before classification, URLs and @-mentions were stripped from the post text. Posts with fewer than 20 alphabetic characters after stripping were treated separately, because language identification on very short texts is unreliable. The langdetect library was seeded deterministically (seed = 0) to ensure reproducible results.

For posts meeting the minimum length, the combination of signals determines the decision group. Table 1 summarises the mapping.

The core group (`core_tag_supported`) contains posts where the author tagged the post as Slovene *and* at least one language identifier agrees. The three identifier-supported review groups extend coverage to posts that were not tagged `sl` but are classified as Slovene by the language identifiers. Posts tagged `sl` with no identifier support (`review_tag_only`) and posts too short for reliable classification (`review_short_tagged`) were excluded after manual validation confirmed their higher noise rate (see Section 4).

Table 1: Decision groups assigned by the language filtering step. Only groups marked as *included* are present in the final corpus.

Decision group	Tag	langid	langdetect	Status
Core tag-supported	yes	yes or no	yes or no	Included
Model consensus only	no	yes	yes	Included
langid only	no	yes	no	Included
langdetect only	no	no	yes	Included
Tag only	yes	no	no	Excluded
Short tagged	yes	not evaluated (text too short)		Excluded

In addition to langid and langdetect, two further language identification tools — fastText (Joulin et al., 2017) and Lingua — were evaluated on the manually validated samples. On the comparable subset, all four detectors showed the same observed precision; the relevant difference was recall. The langid–langdetect pair recovered more acceptable Slovene posts than the alternatives, so the original detector pair was retained for the final corpus build.

3.5 Final corpus assembly

The core and the three included review groups were merged into a single final corpus, deduplicated by post URI. Posts from multiple collection runs (historical crawl, live firehose, and incremental updates) were combined in this step, with the first occurrence of each URI retained. The final corpus contains 141,013 posts.

Each post in the analysis dataset is stored with its text, timestamp, author-supplied language tags, post-type flags (reply, quote), embed kind, and the per-post classifier outputs (decision group, langid label, langdetect probability). No linguistic annotation is included in the present corpus build.

3.6 Unit of analysis

The unit of analysis is the individual post. Language classification is applied per post, not per author or per conversation thread. A post in English by a Slovene-linked author inside an otherwise Slovene thread is classified on its own text and excluded if the text is not Slovene. This means the corpus represents Slovene-language posting, not Slovene-linked authorship in general.

4 VALIDATION

4.1 Precision validation

Validation was carried out through manual annotation of sampled posts. The strict core was evaluated using a random sample of 300 posts drawn uniformly from the `core_tag_supported` group. All 300 sampled posts were judged acceptable as Slovene, yielding an estimated precision of 100% on the core layer.

The broader review layer was evaluated with a separate random sample of 300 posts. In that sample, 214 posts were labelled `Slovene-dominant`, 2 `Mixed-with-Slovene`, 40 `Not-Slovene`, and 44 `Undeterminable/too-short`, for an overall acceptability rate of 72%. A breakdown by decision group revealed that the noise was concentrated in the tag-only and short-tagged subgroups: within this 300-post sample, every row drawn from the `review_model_consensus_only`, `review_langid_only`, and `review_langdetect_only` groups was judged acceptable. Additional dedicated follow-up checks on the two smaller model-supported groups refined that picture: in the `review_langid_only` sample of 50 posts, 49 were labelled `Slovene-dominant` and 1 `Mixed-with-Slovene`; in the `review_langdetect_only` sample of 50 posts, 49 were labelled `Slovene-dominant` and 1 `Not-Slovene`.

These results support a final corpus that includes the strict core and all three model-supported review groups, while excluding the noisier tag-only and short-tagged material.

4.2 False-negative estimation

Precision validation confirms that posts included in the corpus are Slovene. A separate analysis estimated the false-negative rate, i.e. the fraction of Slovene posts from known authors that were excluded by the pipeline. A random sample of 200 posts from seed authors that (a) were not included in the final corpus, (b) carried no Slovene language tag, and (c) met the project's minimum-text rule of 20 alphabetic characters after URL and @-mention stripping was manually annotated. Of these, 17 posts (8.5%) were judged to be Slovene-dominant or Mixed-with-Slovene.

Extrapolating to the full pool of 97,968 excluded untagged posts from known authors that met the same minimum-text rule suggests that roughly 8,300 Slovene posts from the same author set were not captured by the current validated rule. This corresponds to approximately 5.9% of the size of the final corpus. The corpus should therefore be described as a high-precision collection of Slovene posts from discovered Slovene-

linked authors under the current rule, not as a complete census of all Slovene writing on Bluesky.

5 EMPIRICAL ANALYSIS

5.1 Corpus profile

The final corpus contains 141,013 posts by 432 unique authors, spanning 9 August 2023 to 16 April 2026. Table 2 summarises its main properties. Posts are distributed across four decision groups: 85,094 `core_tag_supported` (60.3%), 46,134 `review_model_consensus_only` (32.7%), 5,632 `review_langid_only` (4.0%), and 4,153 `review_langdetect_only` (2.9%).

Table 2: Summary statistics for the final Slovene Bluesky corpus.

Measure	Value
Posts	141,013
Unique authors	432
Date range	9 Aug 2023–16 Apr 2026
Median post length	86 characters
Replies	96,278 (68.3%)
Original posts	41,517 (29.4%)
Non-reply quote posts	3,218 (2.3%)
Authors for 50% of posts	22

5.2 Community formation over time

Monthly metadata were used to distinguish raw post growth from the formation of a more stable author community. For each calendar month, the analysis counted total posts, active authors, first-seen authors, returning authors, cumulative authors, posts per active author, and simple one- and three-month first-seen author retention. All timestamps are treated as UTC. The final month, April 2026, is partial and is therefore excluded from retention windows, monthly period medians, and the statistical comparisons below.

One break stands out clearly in the monthly series. In November 2024, Slovene post volume increased from 31 in October 2024 to 5,984 the following month, while the number of active authors increased from 13 to 196. The number of first-seen authors rose from 8 in October to 172 in November. By the end of November 2024, the cumulative number of observed authors had reached 202, and by the end of December it had reached 252. Figure 1 shows the full monthly post curve.

This break is visible not only in volume but also in participation. Through October 2024, the corpus contains 178 posts by 30 authors, with a median of 10 posts and 3 active authors per complete month. From November 2024 onward, the corpus contains 140,835 posts by 429 authors, with median complete-month values of 7,958 posts and 210 active authors. Median posts per active author also rises from 2.33 before the break to 37.73 after it, indicating that the change is not only an increase in the number of participating authors. Table 3 summarises the period comparison.

Table 3: Community-formation metrics before and after the late-2024 surge. Monthly medians exclude the partial final month, April 2026.

Measure	Through Oct 2024	Nov 2024 onward
Total observed posts	178	140,835
Total active authors	30	429
Median monthly posts	10	7,958
Median monthly active authors	3	210
Median posts per active author	2.33	37.73

Retention by first-seen month points in the same direction: the late-2024 surge does not look like one-off activity alone. Of the 172 authors first seen in November 2024, 86.6% posted again in December 2024 and 94.2% posted at least once in the following three months. Of the 50 authors first seen in December 2024, 72.0% posted again in January 2025 and 82.0% posted at least once in the following three months. These figures do not show why authors became active, but they do suggest that most kept posting.

The late-2024 break occurs well after Bluesky's public opening in February 2024 (Bluesky Social PBC, 2024), suggesting that the removal of invite-only access was a necessary background condition for growth but not, by itself, sufficient to explain the much later Slovene surge. It also coincides with a wider post-election migration wave to Bluesky after the November 2024 U.S. presidential election (Parvini, 2024). These markers are treated as contextual reference points rather than causal explanations.

As a simple robustness check, Mann–Whitney U tests were run on complete monthly observations before and after November 2024. The post-surge months were higher for monthly post counts, active-author counts, and posts per active author in all pairwise comparisons (Cliff's delta = 1.00 in each case; two-sided $p \approx 1.6 \times 10^{-6}$). These tests are used descriptively rather than causally: they support the claim that the late-2024 shift is large and systematic in the observed corpus, but they do not establish its external cause.

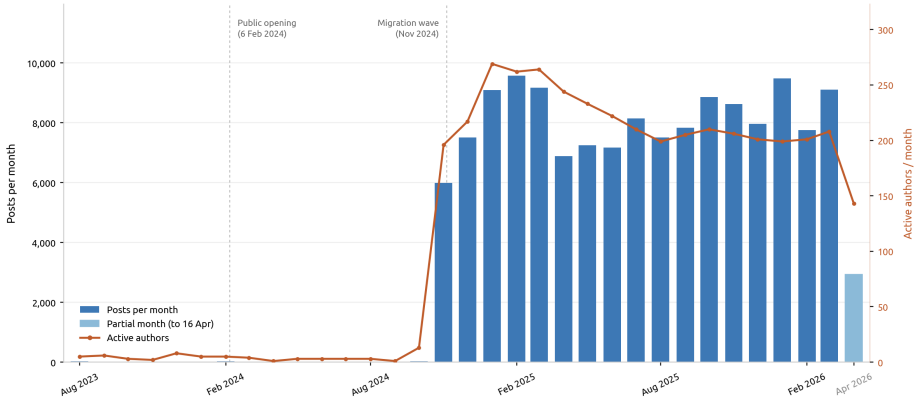

Figure 1: Monthly count of Slovene posts in the final corpus (bars, left axis) and active authors per month (line, right axis), August 2023 to April 2026. Dashed lines mark two contextual reference points: Bluesky's public opening without invitations on 6 February 2024 and the wider post-election migration wave in November 2024. Dotted lines mark the EU Parliament elections (9 June 2024) and the Slovenian parliamentary elections (22 March 2026). The April 2026 bar is partial, ending on 16 April.

5.3 Temporal patterns

Posting activity follows a clear daily rhythm consistent with the Central European time zone. A morning peak occurs at 09:00–10:00 UTC and a stronger evening peak at 18:00–19:00 UTC, corresponding roughly to late morning and evening in Slovenia depending on daylight saving time. Activity is lowest between 01:00 and 04:00 UTC. This pattern is consistent with a community of primarily Slovenian-timezone users.

5.4 Interactional structure

Using a mutually exclusive hierarchy that counts reply-quote combinations as replies, the corpus is dominated by reply posts: 96,278 of 141,013 included posts (68.3%) are replies, while 41,517 (29.4%) are original posts and 3,218 (2.3%) are non-reply quote posts. The median reply is 77 characters, compared to 114 characters for original posts. In this corpus, Slovene Bluesky use is therefore strongly conversational.

This pattern is not only a result of the late-2024 growth surge. Before November 2024, replies already accounted for 54.5% of posts; from November 2024 onward, their share rises to 68.3%. The post-surge influx of authors is therefore consistent with a continuation and strengthening of an interactional pattern that was already present, rather than the introduction of a fundamentally different one.

The author-level distribution tells a similar story. Among the 246 authors with at least 20 posts in the final corpus, the median reply rate is 0.66 (IQR 0.38–0.84), and only 9 authors (3.7%) have a reply rate of zero. At the same time, 74 authors (30.1%) have reply rates above 0.8. This suggests that the dominance of replies is not only an aggregate artefact of corpus-level totals, but a feature that extends beyond a small number of highly active accounts. A simple robustness check leads to the same conclusion: the median monthly reply share is nearly identical in the full corpus and in the subset of authors with at least 20 posts (67.1% vs. 67.3%), indicating that the conversational character of the corpus is not driven mainly by marginal one-off accounts.

A small broadcast-oriented minority is nevertheless visible. Using simple descriptive thresholds based on reply rate, 39 authors in the subset of authors with at least 20 posts fall into a broadcast-heavy group (reply rate below 0.2), 133 into a mixed group, and 74 into a strongly conversational group (reply rate above 0.8). These labels are only descriptive heuristics, not validated social categories, but they help illustrate the broad split already visible in the data: a limited number of accounts contribute mostly non-reply content, while a substantial share of active authors use the platform primarily for exchange and ongoing dialogue.

Posts with emoji occur at a rate of 19.5%, with at least one external URL in the post text at 12.6%, hashtags at 4.9%, and @-mentions at 1.2%. The low hashtag rate is consistent with the conversational character of the corpus. Across the whole corpus, 11.2% of posts embed a structured external link card, 8.0% include images, 2.3% are non-reply quote posts embedding another post, 0.5% contain video, and 0.2% combine a quote-post with attached media. Video is notably rare.

5.5 Author concentration and infrastructure

Posting activity is heavily concentrated. As Table 2 shows, just 22 authors are responsible for half of the corpus. The top 10% of authors (43 users) account for 69.6% of all posts, while the median author contributed 35 posts over the entire collection period. This skewed distribution is characteristic of social media corpora generally, though the concentration here is particularly pronounced given the small and still-forming community.

Regarding platform infrastructure, 423 of 432 authors (97.9%) are hosted on Bluesky's own PDS infrastructure (`*.host.bsky.network`), while 9 (2.1%) are hosted on `eurosky.social`, an independent PDS. This confirms that the corpus is not restricted

to a single server and that DID-based PDS resolution is operational in the collection workflow.

Handle choice is slightly more varied. Of the 432 authors, 395 (91.4%) use a `*.bsky.social` handle, 30 (6.9%) use a user-owned custom domain, and 7 (1.6%) use a `*.eurosky.social` subdomain. The custom-domain group contributed 13,303 posts (9.4% of the corpus). Two authors combine a custom domain handle with `eurosky.social` hosting.

Together, these figures show that most corpus authors use the default Bluesky configuration, while a smaller minority changed either their public handle, their hosting provider, or both.

5.6 Hashtag and linking practices

Hashtag use is sparse throughout the corpus. Only 6,879 of 141,013 posts (4.9%) contain at least one hashtag, suggesting that broad hashtag coordination is not central to this corpus. The most frequent hashtags are often highly concentrated: among the 20 most frequent hashtags, 18 have a single author contributing more than half of all uses. Several of the top hashtags, including `#supslovenija`, `#supdeska`, and `#aquamarina`, are used exclusively by a single account and therefore function more as repeated self-tagging than as markers of a wider community topic. At the same time, a small minority of hashtags shows more distributed authorship, such as `#pohvalanadan` (91 occurrences, 28 authors) and `#ema2025` (67 occurrences, 15 authors), suggesting that wider circulation does occur but is not the dominant pattern.

External link sharing is more prevalent than hashtagging. A total of 17,838 posts (12.6%) contain at least one external domain. The most frequently linked domain is `media.tenor.com` (2,613 occurrences across 121 authors), indicating that GIF-based informal exchange is a prominent and broadly shared interactional practice. Other highly visible domains are more concentrated: `ebx.sh`, a Bluesky-related link-sharing utility, appears 1,804 times but is used by only two authors, while `tehnozvezdje.si` is likewise concentrated in a small number of accounts. These domains point to two different practices: broadly shared informal exchange on the one hand, and repeated account-specific linking on the other.

At the level of domain categories, linked content is shaped primarily by Slovene news media (35.8% of domain occurrences), followed by GIF and media-sharing services (14.5%), platform tooling and Bluesky utilities (11.2%), and video platforms (8.0%). The news category is distributed across several outlets rather than concentrated in a single source, pointing towards recurring current-affairs engagement. At the same

time, the prominence of GIF sharing and Bluesky-native tools suggests a space shaped not only by information sharing, but also by informal exchange and platform-specific habits.

Link-sharing also differs sharply by post type. External links occur in 32.2% of original posts but in only 4.5% of replies. This is consistent with the broader interactional pattern observed in Section 5.4: original posts are comparatively more likely to introduce or circulate content, while replies function primarily as conversational turns. Taken together, the hashtag and domain patterns suggest a small conversational public space oriented around informal exchange and recurring news-media engagement, rather than one organised primarily through hashtag-driven topic publics.

5.7 Linguistic profile and multilinguality

A notable feature of the corpus is the frequency of mixed language tagging. Of the 141,013 posts, 58,416 (41.4%) were tagged with both `sl` and `en` language codes. By contrast, langid detected a non-Slovene primary language in only 7.3% of posts. The predominant non-Slovene labels assigned by langid were Croatian (*hr*: 4,506 posts) and Bosnian (*bs*: 2,285 posts)—closely related South Slavic languages for which cross-label confusion by automatic detectors is common. langid assigned English to only 278 posts.

The high rate of `sl+en` tagging without corresponding English detection by langid suggests that mixed language metadata often reflects client-side tagging behaviour rather than sustained English-dominant content. The corpus nevertheless contains English lexical insertions within otherwise Slovene posts, which are retained as authentic examples of mixed-register writing.

6 DISCUSSION

The empirical findings in the previous section can be read in two ways: as descriptive statistics about a dataset, or as a portrait of a community caught in the process of forming. This section takes the second reading.

6.1 A community with a visible history

Most corpora are static snapshots — the social processes that produced them are invisible in the data. This corpus is unusual in that its own growth structure records a visible transition. The near-zero activity before November 2024 and the 193-fold increase in that month mark the clearest structural break in the corpus. A majority of

authors first appear in the corpus in November or December 2024, while lower-volume activity is already present from August 2023 onward.

This matters for how the corpus is interpreted. The language in it was produced under specific social conditions: during a period of platform disruption, by people establishing or re-establishing an online presence in a small public space. Whether these conditions leave traces in the language is an open question.

6.2 Dialogue over broadcasting

The most striking single finding is the reply rate: 68.3% of all posts are replies, with original posts accounting for 29.4% and non-reply quote posts for 2.3%. This distribution points to use centred more on exchange than on broadcasting.

The author-level evidence points in the same direction. Among the 246 authors with at least 20 posts in the corpus, the median per-author reply rate is 0.66, and only 9 (3.7%) have a reply rate of zero. A robustness check confirms that the median monthly reply share is nearly identical in the full corpus and in this active-author subset (67.1% vs. 67.3%), suggesting that the conversational character of the corpus is not an artefact of marginal one-off contributors. Using simple descriptive thresholds, most active authors fall into a mixed or strongly conversational group; the broadcast-heavy group (reply rate below 0.2) accounts for a small minority of authors and posts. These groupings are heuristic rather than validated social categories, but they help frame the distribution.

Notably, this pattern is not a product of the late-2024 influx of new authors. Before November 2024, replies already accounted for 54.5% of posts in the corpus. The post-surge growth is therefore consistent with a continuation and strengthening of an interactional tendency that was already present, rather than a shift in the kind of activity taking place. Whether this reflects the small-community character of the corpus, platform affordances, or the particular interests of early adopters cannot be established from these data alone.

6.3 Small but coherent

The Slovene Bluesky community captured here is small by the standards of any major language platform, and participation is concentrated around a core group of highly active users. The hashtag and linking patterns make that public space easier to describe more precisely.

Hashtag use is sparse (4.9% of posts) and largely concentrated in individual accounts: 18 of the 20 most frequent hashtags have a single author contributing more than half of all uses. This is consistent with a space that is not organised around broad hashtag-driven topic publics. It is also consistent with the conversational register identified in Section 5.4: when interaction is primarily dialogic, hashtags function less as thematic coordination devices and more as occasional self-tagging or platform-cultural signals.

Linking practices tell a complementary story. External domains appear in 12.6% of posts, more than twice the hashtag rate. The linked content is shaped primarily by Slovene news media (35.8% of domain occurrences), GIF and media-sharing services (14.5%), and Bluesky-native tooling (11.2%), with video platforms (8.0%) also prominent. The news-media linking is distributed across several outlets, suggesting recurring current-affairs engagement rather than dependence on a single source. GIF sharing, by contrast, is a broadly shared informal practice: `media.tenor.com` alone appears in posts by 121 different authors.

Taken as a whole, the hashtag and linking patterns point to a small conversational public space oriented around informal exchange, recurring news-media engagement, and platform-native practices — rather than a thematically organised mass public. The daily rhythm consistent with Slovenian working hours reinforces this portrait of a geographically and linguistically bounded community, though the corpus size makes finer-grained claims about internal subgroups difficult to support.

6.4 ATProto as a research infrastructure

From a methodological standpoint, the experience of building this corpus confirms that ATProto's open design has real consequences for research. Public synchronisation and repository mechanisms made a backfill of the discovered Slovene-linked authors achievable without special research access (Bluesky Social PBC, n.d.-b, n.d.-c). For researchers working on small or under-resourced languages, where corpora are scarce and institutional resources are limited, this openness matters. The Slovene community on Bluesky may be small, but the corpus captures much of the community's publicly available posting history.

7 LIMITATIONS AND ETHICS

7.1 Coverage limits

The corpus should not be described as a complete census of all Slovene Bluesky posts ever published. It covers currently recoverable public posts from discovered Slovene-

linked accounts; deleted posts are excluded by design. The false-negative analysis (Section 4) estimates an 8.5% miss rate within the pool of excluded untagged posts from known authors that met the project's minimum-text rule; the true miss rate across the full user population is unknown and likely higher.

The methodology uses post-level rather than thread-level language classification. Slovene-linked users may post in English or other languages within otherwise Slovene conversations, and such replies are not automatically treated as Slovene corpus material.

7.2 Open data and reproducibility

The AT Protocol's openness is what made this corpus possible. The collection described here used public protocol mechanisms rather than a paid or application-only research API. This does not mean that access is unlimited: Bluesky documents rate limits for its services, and the ATProto sync specification explicitly allows service providers to apply rate limits and data quotas (Bluesky PBC, n.d.-c; Bluesky Social PBC, n.d.-c). For a small-language community like Slovene, however, the availability of public synchronisation mechanisms is still a practically valuable property, though it also raises specific questions about redistribution.

7.3 Privacy and identifiability

The same openness that makes collection easy raises a concern worth stating plainly. The Slovene Bluesky community, at the time of collection, consists of 432 identified authors — a small number. In most cases, the individuals behind the accounts can be identified with little effort: handles are publicly visible, many users link their real names or professional identities through custom domains, and the posting history of any individual is freely accessible through DID resolution. The corpus therefore represents not just a dataset but a detailed longitudinal record of specific, identifiable people.

This is not unique to Bluesky or to this corpus — it is a structural feature of public social-media research. *Public* does not mean *anonymous*: social-media content can remain identifiable even when obvious identifiers are removed, and direct quotation or full-text redistribution can make reverse identification easier (Ayers et al., 2018; Franzke et al., 2019). Users who post under their real names may not have anticipated that their posting history would be compiled into a citable research corpus. Following standard practice, full post texts are not reproduced in this article, and any future redistribution should take a conservative form to limit re-identification risk.

7.4 Corpus redistribution on a decentralised platform

Whether and how to redistribute this corpus is not straightforward, and this paper does not attempt to resolve the question. ATProto supports content and account deletion, and its synchronisation specification explicitly warns against public redistribution of static repository snapshots in bulk, as mirrors are expected to respect record deletions and account-status changes (Bluesky Social PBC, n.d.-c). A static corpus distributed as a file cannot fully reflect this behaviour: posts deleted on Bluesky remain present in copies that have already been downloaded elsewhere. From a European data protection perspective, public posts are still considered personal data when authors are identifiable (European Parliament and Council of the European Union, 2016).

Identifier-based redistribution has often been used in Twitter research to balance reproducibility with deletion and access constraints (Kinder-Kurlanda et al., 2017). For ATProto, an equivalent and widely accepted redistribution model has not yet emerged. In the present study, the corpus is therefore described but not released as a full-text dataset. The methodology and collection code are made available, and it is in principle possible to reassemble the corpus from public ATProto data under the conditions described in this paper.

At the same time, it is the author's intention to make the corpus available in a form that is legally and ethically defensible. A carefully defined redistribution model would enable reuse without requiring individual researchers to repeat the full collection process on their own. However, determining an appropriate access format for a small and identifiable language community requires further consideration and is treated as future work rather than resolved here.

These issues are noted to document the methodological limits of working with public data on open-protocol platforms, especially in small language communities.

8 FUTURE WORK

The most immediate next step is a legally and ethically defensible access format. The current public material consists of methodology and collection code only. A fuller corpus release would make the resource usable without re-running the entire pipeline, which matters for researchers without ATProto collection experience, but it should wait until the redistribution questions discussed in Section 7 have been addressed. Possible formats include stable identifiers, strongly pseudonymised text, or controlled access rather than an unrestricted full-text dump.

A further step would be structured community consultation. Because this corpus captures a small and identifiable public, technically valid release formats may still differ in how acceptable they feel to the users whose posts are included. Future work could therefore combine legal and methodological analysis with lightweight community consultation on corpus access, quotation, and redistribution practices.

Adding linguistic annotation would extend the corpus considerably. A candidate pipeline has been prepared: CLASSLA for sentence segmentation, tokenisation, lemmatisation, and Slovene morphosyntactic tags (Ljubešić & Dobrovoljc, 2019), with Trankit providing Universal Dependencies annotation (Nguyen et al., 2021). This follows the broad direction established by the JANES project for Slovene user-generated content (Erjavec et al., 2019). The pipeline has not yet been run over the full corpus due to computational cost.

The corpus can also support follow-up work on non-standard Slovene processing, code-switching between Slovene and neighbouring languages, thread-level dialogue, and community formation during platform migration. The high reply rate (68.3%) makes thread reconstruction especially relevant, while the temporal structure makes the corpus useful for longitudinal study.

Finally, the collection infrastructure can be extended longitudinally and applied to other small-language communities on Bluesky. That would make it possible to compare how under-resourced languages establish themselves on an open-protocol platform, and to test whether the Slovene pattern reported here is exceptional or part of a broader ATProto trajectory.

9 CONCLUSION

This paper has described a workflow for building a high-precision corpus of public Slovene Bluesky posts and used that corpus for an initial look at the interactional and temporal structure of Slovene activity on the platform.

The pipeline combines relay-based tagged discovery (historical crawl and live firehose), DID-based backfill from individual Personal Data Servers, and post-level language filtering validated through manual annotation. It is fully protocol-native: no special API keys, no institutional data-access agreements, and no restriction to a single hosting server. For small-language communities where research infrastructure is limited, this openness is a practical advantage.

The corpus of 141,013 posts by 432 authors shows a community whose recoverable activity expands sharply in late 2024, is dominated by dialogue rather than broad-

casting, and is concentrated around a small core of highly active users. The conversational character predates the late-2024 growth surge and holds across much of the active author base, not only in aggregate totals. Hashtag use is sparse and largely account-concentrated, while linked content is shaped by a mix of Slovene news media, GIF-based informal exchange, and platform-native tools — a pattern consistent with a small conversational public space organised around dialogue and recurring content-sharing rather than broad topic publics. The small size of the community makes finer-grained claims about internal subgroups difficult to support, but the combination of temporal, interactional, and linking evidence points towards a geographically and linguistically bounded community with a recognisable overall profile.

This corpus should be treated as a snapshot of currently recoverable public posts from discovered Slovene-linked authors, not a complete census of all Slovene writing on Bluesky. Deleted posts are excluded by design, and within the pool of excluded untagged posts from known authors that meet the project's minimum-text rule, the false-negative rate is estimated at approximately 8.5%. The main next step is not only technical enrichment, but a redistribution model that makes the corpus useful without ignoring the rights and expectations of the people whose posts make it possible.

10 DATA AND CODE AVAILABILITY

The collection and anonymisation code is available at https://github.com/hulln/slosky. The corpus itself is not distributed as a full-text release at this stage; see Sections 7 and 8. The corpus can be reconstructed from currently public ATProto data using the methodology and code described in this paper.

ACKNOWLEDGEMENTS

Generative AI tools were used during the coding, implementation, analysis support, and writing phases of this paper, primarily ChatGPT/Codex (GPT-5.4) and Claude Code (Claude Sonnet 4.6). The research topic, overall direction of the work, and final interpretive decisions were determined by the author. All generated material was reviewed, revised where necessary, and finalised by the author.

REFERENCES

Ayers, J. W., Caputi, T. L., Nebeker, C., & Dredze, M. (2018). Don't quote me: Reverse identification of research participants in social media studies. *NPJ Digital Medicine*, *1*(1), 30. https://doi.org/10.1038/s41746-018-0036-2

Balduf, L., Sokoto, S., Baronchelli, A., Castro, I., Król, M., Tyson, G., Pavlou, G., Scheuermann, B., & Ascigil, O. (2025). Bootstrapping social networks: Lessons from Bluesky starter packs. *Proceedings of the Nineteenth International AAAI Conference on Web and Social Media*, *19*, 178–192. https://doi.org/10.1609/icwsm.v19i1.35810

Bluesky PBC. (n.d.-a). *com.atproto.sync.listReposByCollection*. Retrieved April 21, 2026, from https://docs.bsky.app/docs/api/com-atproto-sync-list-repos-by-collection

Bluesky PBC. (n.d.-b). *Posts*. Retrieved April 21, 2026, from https://docs.bsky.app/docs/advanced-guides/posts

Bluesky PBC. (n.d.-c). *Rate limits*. Retrieved April 21, 2026, from https://docs.bsky.app/docs/advanced-guides/rate-limits

Bluesky Social PBC. (n.d.-a). *Repository*. Retrieved April 21, 2026, from https://atproto.com/specs/repository

Bluesky Social PBC. (n.d.-b). *Streaming data*. Retrieved April 21, 2026, from https://atproto.com/guides/streaming-data

Bluesky Social PBC. (n.d.-c). *Sync*. Retrieved April 21, 2026, from https://atproto.com/specs/sync

Bluesky Social PBC. (2023a). The AT protocol [Technical specification]. Retrieved April 1, 2026, from https://atproto.com/guides/overview

Bluesky Social PBC. (2023b). *Bluesky private beta update*. Retrieved April 22, 2026, from https://bsky.social/about/blog/3-2-2023-bluesky-beta-app

Bluesky Social PBC. (2023c). *Private beta update & roadmap*. Retrieved April 22, 2026, from https://bsky.social/about/blog/6-02-2023-beta-update

Bluesky Social PBC. (2024). *Join Bluesky today (bye, invites!)* Retrieved April 22, 2026, from https://bsky.social/about/blog/02-06-2024-join-bluesky

Erjavec, T., Fišer, D., Čibej, J., Arhar Holdt, Š., Ljubešić, N., Dobrovoljc, K., & Krek, S. (2019). The Janes project: Language resources and tools for Slovene user generated content. *Language Resources and Evaluation*, *53*, 223–246. https://doi.org/10.1007/s10579-018-9425-z

European Parliament and Council of the European Union. (2016). Regulation (EU) 2016/679 of the European Parliament and of the Council of 27 April 2016 on the protection of natural persons with regard to the processing of personal data and on the free movement of such data, and repealing Directive 95/46/EC (General Data Protection Regulation). Retrieved April 21, 2026, from https://eur-lex.europa.eu/eli/reg/2016/679/oj/eng

Failla, A., & Rossetti, G. (2024). "I'm in the Bluesky Tonight": Insights from a year worth of social data. *PLOS ONE*, *19*(11), e0310330. https://doi.org/10.1371/journal.pone.0310330

Franzke, A. S., Bechmann, A., Zimmer, M., Ess, C., & Association of Internet Researchers. (2019). Internet research: Ethical guidelines 3.0. Retrieved April 21, 2026, from https://aoir.org/reports/ethics3.pdf

Gorjanc, V., & Krek, S. (Eds.). (2005). *Študije o korpusnem jezikoslovju*.

Hanny, D., Schmidt, S., & Resch, B. (2026). Bluesky as a social media data source for disaster management: Investigating spatio-temporal, semantic and emotional patterns for floods and wildfires. *Journal of Computational Social Science*, *9*, 16. https://doi.org/10.1007/s42001-025-00448-x

Joulin, A., Grave, E., Bojanowski, P., & Mikolov, T. (2017). Bag of tricks for efficient text classification. *Proceedings of the 15th Conference of the European Chapter of the Association for Computational Linguistics: Volume 2, Short Papers*, 427–431. https://aclanthology.org/E17-2068/

Kinder-Kurlanda, K., Weller, K., Zenk-Möltgen, W., Pfeffer, J., & Morstatter, F. (2017). Archiving information from geotagged tweets to promote reproducibility and comparability in social media research. *Big Data & Society*, *4*(2). https://doi.org/10.1177/2053951717736336

Kleppmann, M., Frazee, P., Gold, J., Graber, J., Holmgren, D., Ivy, D., Johnson, J., Newbold, B., & Volpert, J. (2024). Bluesky and the AT protocol: Usable decentralized social media. *Proceedings of the ACM CoNEXT-2024 Workshop on the Decentralization of the Internet*, 1–7. https://doi.org/10.1145/3694809.3700740

Krek, S., Arhar Holdt, Š., Erjavec, T., Čibej, J., Repar, A., Gantar, P., Ljubešić, N., Kosem, I., & Dobrovoljc, K. (2020a). Gigafida 2.0: The reference corpus of written standard Slovene. In N. Calzolari, F. Béchet, P. Blache, K. Choukri, C. Cieri, T. Declerck, S. Goggi, H. Isahara, B. Maegaard, J. Mariani, H. Mazo, A. Moreno, J. Odijk, & S. Piperidis (Eds.), *Proceedings of the twelfth language resources and evaluation conference* (pp. 3340–3345). European Language Resources Association. https://aclanthology.org/2020.lrec-1.409/

Krek, S., Erjavec, T., Dobrovoljc, K., Gantar, P., Arhar Holdt, Š., Čibej, J., & Brank, J. (2020b). The ssj500k training corpus for slovene language processing. In D. Fišer & T. Erjavec (Eds.), *Jezikovne tehnologije in digitalna humanistika: Zbornik konference* (pp. 24–33). Inštitut za novejšo zgodovino. http://nl.ijs.si/jtdh20/pdf/JT-DH_2020_Krek-et-al_The-ssj500k-Training-Corpus-for-Slovene--Language-Processing.pdf

Krek, S., Erjavec, T., Repar, A., Čibej, J., Arhar Holdt, Š., Gantar, P., Kosem, I., Robnik Šikonja, M., Ljubešić, N., Dobrovoljc, K., Laskowski, C., Grčar, M., Holozan, P., Šuster, S., Gorjanc, V., Stabej, M., & Logar, N. (2019). *Corpus of written standard Slovene Gigafida 2.0*. Slovenian language resource repository CLARIN.SI. http://hdl.handle.net/11356/1320

Ljubešić, N., & Dobrovoljc, K. (2019). What does neural bring? analysing improvements in morphosyntactic annotation and lemmatisation of Slovenian, Croatian and Serbian. *Proceedings of the 7th Workshop on Balto-Slavic Natural Language Processing*, 29–34. https://doi.org/10.18653/v1/W19-3704

Lui, M., & Baldwin, T. (2012). Langid.py: An off-the-shelf language identification tool. *Proceedings of the ACL 2012 System Demonstrations*, 25–30. https://aclanthology.org/P12-3005/

Nakatani, S. (2010). Language detection library for Java [Software library]. https://github.com/shuyo/language-detection

Nguyen, M. V., Lai, V. D., Pouran Ben Veyseh, A., & Nguyen, T. H. (2021). Trankit: A light-weight transformer-based toolkit for multilingual natural language processing. *Proceedings of the 16th Conference of the European Chapter of the Association for Computational Linguistics: System Demonstrations*, 80–90. https://doi.org/10.18653/v1/2021.eacl-demos.10

Pančur, A., Erjavec, T., Meden, K., Ojsteršek, M., Šorn, M., & Blaj Hribar, N. (2022). *Slovenian parliamentary corpus (1990-2022) siParl 3.0*. Slovenian language resource repository CLARIN.SI. http://hdl.handle.net/11356/1748

Parvini, S. (2024). *Bluesky has added 1 million users since the US election as people seek alternatives to X*. Retrieved April 22, 2026, from https://apnews.com/article/d78de194e0e85f803652b70b161a963b

Quelle, D., & Bovet, A. (2025). Bluesky: Network topology, polarization, and algorithmic curation. *PLOS ONE*, *20*(2), e0318034. https://doi.org/10.1371/journal.pone.0318034

Tjong Kim Sang, E. (2025). Using Bluesky for social media analysis. *Digital Humanities Benelux 2025*. https://doi.org/10.5281/zenodo.15650889

SKUPNOST V NASTAJANJU: GRADNJA IN ANALIZA KORPUSA SLOVENSKIH OBJAV NA PLATFORMI BLUESKY

Prispevek predstavlja korpus 141.013 javnih objav v slovenščini, ki jih je na mikroblogerski platformi Bluesky med avgustom 2023 in sredino aprila 2026 objavilo 432 avtorjev. Korpus je bil oblikovan z večstopenjskim postopkom na ravni protokola AT, ki zajema odkrivanje objav z oznako *sl*, rekonstrukcijo javno dostopnih zgodovin objav posameznih računov ter filtriranje posameznih objav glede na jezik. Kot ena prvih empiričnih, jezikovno specifičnih študij rabe platforme Bluesky za slovensko jezikovno skupnost se prispevek osredotoča na oblikovanje novega, na odprtem protokolu temelječega javnega prostora. Na podlagi zbranih podatkov je opazen izrazit porast aktivnosti konec leta 2024. Odgovori na objave predstavljajo 68,3 % vseh objav, pri čemer je njihov delež visok že pred porastom aktivnosti. Opazna je tudi redka in avtorsko koncentrirana raba ključnikov, večinoma s slovenskimi mediji povezano deljenje novic ter neformalna raba GIF-ov. Prispevek pokaže, kako lahko dostop na ravni protokola AT omogoča gradnjo korpusa za jezike z manjšim številom govorcev, hkrati pa odpira pravna in etična vprašanja, povezana z dostopnostjo javnih podatkov manjših jezikovnih skupnosti.

Opomba k jeziku: Angleščina je v tem prispevku uporabljena izključno z namenom nagovarjanja širše mednarodne raziskovalne skupnosti na področjih digitalne humanistike, obdelave naravnega jezika in ATProta. Uporaba angleščine nikakor ne zmanjšuje pomena slovenščine v raziskovalnem in spletnem prostoru, katere vlogo še naprej dosledno in odločno zagovarjam.

Ključne besede: Bluesky, ATProto, slovenščina, korpus, družbena omrežja

To delo je ponujeno pod licenco Creative Commons: Priznanje avtorstva-Deljenje pod enakimi pogoji 4.0 Mednarodna.

This work is licensed under the Creative Commons Attribution-ShareAlike 4.0 International.

https://creativecommons.org/licenses/by-sa/4.0/